

NISHAT RAIHAN

Provost's Postdoctoral Fellow, University of Notre Dame

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No H-1B visa sponsorship required

APPOINTMENTS

2026 – 2028
Notre Dame, IN
Provost's Postdoctoral Fellow
University of Notre Dame
Department of Computer Science and Engineering
Faculty mentor: Dr. Joanna C. S. Santos

EDUCATION

2021 – 2026
Fairfax, VA
Ph.D. in Computer Science
George Mason University · CGPA: 3.96/4.00
Advisors: Dr. Marcos Zampieri and Dr. Antonios Anastasopoulos

2021 – 2024
Fairfax, VA
M.S. in Computer Science
George Mason University · CGPA: 3.97/4.00

2015 – 2018
Gazipur,
Bangladesh
B.Sc. in Computer Science and Engineering
Islamic University of Technology · CGPA: 3.70/4.00

PUBLICATIONS

40+ publications · 650+ citations · h-index 14 · i10-index 19 · Google Scholar

PEER-REVIEWED CONFERENCE PAPERS

- [C12] **Nishat Raihan**, Antonios Anastasopoulos, Marcos Zampieri. "TigerCoder: A Novel Suite of LLMs for Code Generation in Bangla." *Proceedings of the Language Resources and Evaluation Conference (LREC 2026)*. [PDF]
Code LLMs
- [C11] **Nishat Raihan**, Noah Erdachew^m, Jayoti Devi^m, Joanna C. S. Santos, Marcos Zampieri. "CodeGuard: Improving LLM Guardrails in CS Education." *Findings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (EACL 2026)*. [PDF]
AI Safety
- [C10] **Nishat Raihan**, Marcos Zampieri. "TigerLLM: A Family of Bangla Large Language Models." *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025)*, Short Papers. [PDF]
NLP
- [C9] **Nishat Raihan**, Antonios Anastasopoulos, Marcos Zampieri. "mHumanEval: A Multilingual Benchmark to Evaluate Large Language Models for Code Generation." *Proceedings of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL 2025)*, Main, Long Papers. [PDF]
Code LLMs
- [C8] **Nishat Raihan**, Joanna C. S. Santos, Marcos Zampieri. "MojoBench: Language Modeling and Benchmarks for Mojo." *Findings of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL 2025)*. [PDF]
Code LLMs

- [C7] **Nishat Raihan**, Mohammed Latif Siddiq, Joanna C. S. Santos, Marcos Zampieri. "Large Language Models in Computer Science Education: A Systematic Literature Review." *Proceedings of the 56th ACM Technical Symposium on Computer Science Education (SIGCSE-TS 2025)*. [PDF]
CS Education
- [C6] **Nishat Raihan**, Christian Newman, Marcos Zampieri. "Code LLMs: A Taxonomy-based Survey." *2024 IEEE International Conference on Big Data (IEEE BigData 2024)*. [PDF]
Code LLMs
- [C5] **Nishat Raihan**, Dhiman Goswami, Sadiya Sayara Chowdhury Puspo, Christian Newman, Tharindu Ranasinghe, Marcos Zampieri. "CSEPrompts: A Benchmark of Introductory Computer Science Prompts." *27th International Symposium on Methodologies for Intelligent Systems (ISMIS 2024)*. [PDF]
CS Education
- [C4] **Md Nishat Raihan**, Sadiya Sayara Chowdhury Puspo, Shafkat Farabi, Ana-Maria Bucur, Tharindu Ranasinghe, Marcos Zampieri. "MentalHelp: A Multi-Task Dataset for Mental Health in Social Media." *2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*. [PDF]
NLP
- [C3] **M. Raihan**, N. Ulfat, N. Saqib. "An Experimental Analysis on the Sensitivity of the Most Widely Used Edge Detection Methods to Different Noise Types." *International Conference on Computing Advancements (ICCA 2022)*. [PDF]
Vision
- [C2] T. Ahmed, A. Rahman, T. M. Chowdhury, R. Kushol, **M. N. Raihan**. "A Novel Approach to Classify Electrocardiogram Signals Using Deep Neural Networks." *2nd International Conference on Computer and Information Sciences (ICCIS 2020)*, IEEE. [PDF]
Time Series
- [C1] T. Ahmed, **M. N. Raihan**, R. Kushol, M. S. Salekin. "A Complete Bangla Optical Character Recognition System: An Effective Approach." *22nd International Conference on Computer and Information Technology (ICCIT 2019)*, IEEE. [PDF]
Vision

JOURNAL ARTICLES

- [J2] **Nishat Raihan**, Dhiman Goswami, Sadiya Sayara Chowdhury Puspo, Mohammed Latif Siddiq, Christian Newman, Tharindu Ranasinghe, Joanna C. S. Santos, Marcos Zampieri. "On the Performance of Large Language Models on Introductory Programming Assignments." *Journal of Intelligent Information Systems, Springer*, 2025. [PDF]
Code LLMs
- [J1] Rafsanjany Kushol, Imamul Ahsan, **Md. Nishat Raihan**. "An Android-Based Useful Text Extraction Framework Using Image and Natural Language Processing." *International Journal of Computer Theory and Engineering (IJCTE)*, vol. 10, no. 3, pp. 77–83, 2018. [PDF]
Vision

WORKSHOP AND SHARED-TASK PAPERS

- [W18] **Nishat Raihan**, Marcos Zampieri. "Temporal Text Classification with Large Language Models." *Proceedings of the 6th International Conference on Natural Language Processing for the Digital Humanities (NLP4DH 2026)*. [PDF]
NLP
- [W17] **Nishat Raihan**, Sadiya Sayara Chowdhury Puspo, Ana-Maria Bucur, Stevie Chancellor, Marcos Zampieri. "Large Language Models for Mental Health: A Multilingual Evaluation." *2nd Workshop on Language Models for Low-Resource Languages (LoResLM @ EAACL 2026)*. [PDF]
NLP

- [W16] **Nishat Raihan**, Mohammad Anas Jawad, Md Mezbaour Rahman, Noshin Ulfat, Pranav Gupta, Mehrab Mustafy Rahman, Shubhra Kanti Karmakar, Marcos Zampieri. "Overview of BLP-2025 Task 2: Code Generation in Bangla." *Proceedings of the Second Workshop on Bangla Language Processing (BLP @ IJCNLP-AAACL 2025)*. [PDF]
Code LLMs
- [W15] Wren McQueary, Sadia Afrin Mim, **Md. Nishat Raihan**, Justin Smith, Brittany Johnson. "Py-holmes: Causal Testing for Deep Neural Networks in Python." *Companion Proceedings of the 32nd ACM International Conference on the Foundations of Software Engineering (FSE 2024)*. [PDF]
ML Testing
- [W14] Matthew Shardlow et al. (incl. **M. N. Raihan**). "The BEA 2024 Shared Task on the Multilingual Lexical Simplification Pipeline." *19th Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2024)*. [PDF]
NLP
- [W13] Matthew Shardlow et al. (incl. **M. N. Raihan**). "An Extensible Massively Multilingual Lexical Simplification Pipeline Dataset Using the MultiLS Framework." *3rd Workshop on Tools and Resources for People with Reading Difficulties (READI @ LREC-COLING 2024)*. [PDF]
NLP
- [W12] **Nishat Raihan** et al. "EmoMix-3L: A Code-Mixed Dataset for Bangla-English-Hindi Emotion Detection." *WILDRE Workshop, LREC-COLING 2024*. [PDF]
NLP
- [W11] **Md Nishat Raihan**, Dhiman Goswami, Al Nahian Bin Emran, Sadiya Sayara Chowdhury Puspo, Amrita Ganguly, Marcos Zampieri. "MasonTigers at SemEval-2024 Task 9: Solving Puzzles with an Ensemble of Chain-of-Thoughts." *18th International Workshop on Semantic Evaluation (SemEval @ NAACL 2024)*. [PDF]
NLP
- [W10] D. Goswami, S. S. C. Puspo, **M. N. Raihan**, A. N. B. Emran, A. Ganguly, M. Zampieri. "MasonTigers at SemEval-2024 Task 1: An Ensemble Approach for Semantic Textual Relatedness." *18th International Workshop on Semantic Evaluation (SemEval @ NAACL 2024)*. [PDF]
NLP
- [W9] S. S. C. Puspo, **M. N. Raihan**, D. Goswami, A. N. B. Emran, A. Ganguly, O. Uzuner. "MasonTigers at SemEval-2024 Task 8: Performance Analysis of Transformer-based Models on Machine-Generated Text Detection." *18th International Workshop on Semantic Evaluation (SemEval @ NAACL 2024)*. [PDF]
AI Safety
- [W8] A. Ganguly, S. S. C. Puspo, D. Goswami, **M. N. Raihan**. "MasonPerplexity at ClimateActivism 2024: Integrating Advanced Ensemble Techniques and Data Augmentation for Climate Activism Stance and Hate Event Identification." *LT-EDI Workshop, EACL 2024*. [PDF]
NLP
- [W7] A. Ganguly, A. N. B. Emran, S. S. C. Puspo, **M. N. Raihan**, D. Goswami, M. Zampieri. "MasonPerplexity at Multimodal Hate Speech Event Detection 2024: Hate Speech and Target Detection Using Transformer Ensembles." *7th Workshop on Challenges and Applications of Automated Extraction of Socio-political Events from Text (CASE @ EACL 2024)*. [PDF]
NLP
- [W6] D. Goswami, S. S. C. Puspo, **M. N. Raihan**, A. N. B. Emran. "MasonTigers at LT-EDI-2024: An Ensemble Approach towards Detecting Homophobia and Transphobia in Social Media Comments." *4th Workshop on Language Technology for Equality, Diversity, Inclusion (LT-EDI @ EACL 2024)*. [PDF]
NLP

- [W5] **Md Nishat Raihan**, Umma Hani Tanmoy, Anika Binte Islam, Kai North, Tharindu Ranasinghe, Antonios Anastasopoulos, Marcos Zampieri. "Offensive Language Identification in Transliterated and Code-Mixed Bangla." *1st International Workshop on Bangla Language Processing (BLP @ EMNLP 2023)*, Singapore. [PDF] [Dataset]
NLP Best Paper Award
- [W4] **Md Nishat Raihan**, Dhiman Goswami, Sadiya Sayara Chowdhury Puspo, Marcos Zampieri. "nlpBDpatriots at BLP-2023 Task 1: A Two-Step Classification for Violence Inciting Text Detection in Bangla." *1st International Workshop on Bangla Language Processing (BLP @ EMNLP 2023)*, Singapore. [PDF]
NLP
- [W3] Dhiman Goswami, **Md Nishat Raihan**, Sadiya Sayara Chowdhury Puspo, Marcos Zampieri. "nlpBDpatriots at BLP-2023 Task 2: A Transfer Learning Approach to Bangla Sentiment Analysis." *1st International Workshop on Bangla Language Processing (BLP @ EMNLP 2023)*, Singapore. [PDF]
NLP
- [W2] Dhiman Goswami, **Md Nishat Raihan**, Antara Mahmud, Antonios Anastasopoulos, Marcos Zampieri. "OffMix-3L: A Novel Code-Mixed Dataset in Bangla-English-Hindi for Offensive Language Identification." *1st Workshop in South East Asian Language Processing (SEALP @ AACL 2023)*, Bali, Indonesia. [PDF] [Dataset]
NLP
- [W1] **Md Nishat Raihan**, Dhiman Goswami, Antara Mahmud, Antonios Anastasopoulos, Marcos Zampieri. "SentMix-3L: A Bangla-English-Hindi Code-Mixed Dataset for Sentiment Analysis." *11th International Workshop on Natural Language Processing for Social Media (AACL 2023)*, Bali, Indonesia. [PDF] [Dataset]
NLP

AWARDS AND HONORS

2026 – 2028	Provost's Postdoctoral Fellowship, College of Engineering, University of Notre Dame
2026	Doctoral Research Scholarship (Tier 1), The Graduate Division, George Mason University (declined)
2026	Distinguished Academic Achievement Award (Ph.D.), George Mason University
2024	Distinguished Academic Achievement Award (M.S.), George Mason University
2023	Best Paper Award, 1st Workshop on Bangla Language Processing (BLP @ EMNLP 2023)

INVITED TALKS

May 2026	California State University, Fresno Title: Adapting Code LLMs for Under-Explored Domains
May 2026	Prairie View A&M University Title: Adapting Code LLMs for Under-Explored Domains
Apr. 2026	Duke University Title: Adapting Code LLMs for Under-Explored Domains

RESEARCH EXPERIENCE

2026 – 2028
Notre Dame, IN

Provost's Postdoctoral Fellow

Department of Computer Science and Engineering, University of Notre Dame

- Research on safety-by-construction guardrails and multilingual program synthesis for code LLMs, mentored by Dr. Joanna C. S. Santos.

2024 – 2026
Fairfax, VA

Graduate Research Assistant

Department of Computer Science, George Mason University

- Led end-to-end LLM development: data curation, pretraining, fine-tuning, and evaluation for code generation and multilingual settings.
- Released open benchmarks, corpora, and models (mHumanEval, TigerLLM, TigerCoder, MojoBench) adopted by the research community.
- Authored and co-authored peer-reviewed papers, presented at international venues, and collaborated across NLP and software engineering groups.

TEACHING EXPERIENCE

2021 – 2024
Fairfax, VA

Graduate Teaching Assistant (8 semesters)

Department of Computer Science, George Mason University

- CS 583: Analysis of Algorithms (graduate); led labs for sections of 75–90 MS and Ph.D. students.
- CS 262: Introduction to Low-Level Programming (12 labs, 3 projects, midterm, final).
- CS 222: Computer Programming for Engineers / Introduction to Programming.
- Designed inquiry-based lab exercises, including debugging labs built around flawed Quicksort implementations and AI code-review activities in which students critique LLM-generated code.

2019
Dhaka,
Bangladesh

Lecturer, Computer Science

Uttara University

- Taught theory and lab courses: Introduction to Computers, C, and C++.

PROPOSED COURSE

Responsible AI Deployment An upper-level elective bridging foundational algorithms and AI-augmented practice: LLM-assisted code generation, debugging, and responsible deployment.

GRANT AND PROPOSAL EXPERIENCE

Research supported in part by the Naval Sea Systems Command (NAVSEA) and multiple STTR awards, with hands-on exposure to proposal development, budget planning, and articulating measurable research impact.

MENTORING

- **Manmit Singh** · Ph.D. student (2nd year), Duke University.
- **Dipak Meher** · Ph.D. student (3rd year), George Mason University.
- **Jayoti Devi** · Ph.D. student (1st year), George Mason University.
- **Noah Erdachew** · Senior undergraduate, University of Notre Dame.

PROFESSIONAL SERVICE

ORGANIZING

2025	Organizer, BLP-2025 Shared Task on Code Generation in Bangla, the first shared task in this area, with 158 international participants
2025	Organizing Committee, 1st Workshop on Language Models for Low-Resource Languages (LoResLM @ COLING 2025)
2023, 2025	Organizing Committee, 1st and 2nd Workshops on Bangla Language Processing (BLP @ EMNLP 2023; BLP @ IJCNLP-AAACL 2025)
2024	Organizing Committee, Multilingual Lexical Simplification Pipeline (MLSP) Shared Task
2025	Organizing Committee, 3rd International Conference on Foundation and Large Language Models

REVIEWING

ACL Rolling Review (every cycle, 2024 – present) · SIGCSE Technical Symposium (2024 – 2026) · LREC (2026) · COLING (2024 – 2025) · SemEval (2024)

INDUSTRY EXPERIENCE

2019 – 2021 Dhaka, Bangladesh	Software Engineer, RPA and OCR Development Samsung R&D Institute Bangladesh Built production-grade RPA bots and services (Automation Anywhere, UiPath) for international organizations; integrated OCR engines (Tesseract 4, Google Vision, ABBYY, Azure, Tegaki) into IQ Bots with custom Python pipelines. Platforms: Automation Anywhere, UiPath, OpenText, SAP.
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LEADERSHIP AND OUTREACH

2023 – 2024	Vice-President, Bangladesh Graduate Student Association (BDGSA), George Mason University
2022 – 2023	Director of Sports, Bangladesh Graduate Student Association (BDGSA), George Mason University

REFERENCES

Available upon request.